

Place Value with Base-Ten Models

Answer Key

Grade 2 Math

Quick Answers

- | | |
|-------------------------------------|-------------------------------------|
| 1. 135 | 5. 80 |
| 2. 246 | 6. $>$ |
| 3. 308 | 7. hundreds = 4, tens = 0, ones = 7 |
| 4. hundreds = 5, tens = 7, ones = 3 | 8. the number = 125, — |
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What is verified

7 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 2 Math

2.NBT.A – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–4 of 5 across 13 tagged checks.

Common wrong answers

3. *If they got 38: ignored the empty tens place and pushed the digits together*
5. *If they got 8: gave the digit itself instead of its value*
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1. Write the number the model shows. (1 flat, 3 rods, 5 cubes)

Solution: The model shows 1 flat (1 hundred), 3 rods (3 tens), and 5 cubes (5 ones). That is $100 + 30 + 5$. 135

2. A model shows 2 hundreds, 4 tens, and 6 ones. Write the number.

Solution: 2 hundreds is 200, 4 tens is 40, 6 ones is 6. Together: $200 + 40 + 6$. 246

3. Write the number the model shows. (3 flats, no rods, 8 cubes)

Solution: The model has 3 flats (3 hundreds) and 8 cubes (8 ones) — and *no* rods at all, so the tens place holds a zero. The number is $300 + 0 + 8$. Watch for the zero: the answer is not 38. 308

4. Fill in the blanks for the number 573.

Solution: Read each digit by its place: the 5 sits in the hundreds place, the 7 in the tens place, and the 3 in the ones place. So 573 has

hundreds,

tens, and

ones.

5

7

3

5. What is the value of the tens digit in 482?

Solution: The tens digit of 482 is 8, and a digit in the tens place is worth ten times itself: $8 \times 10 = 80$. The digit is 8, but its *value* is

80

6. Model A and Model B: write $<$, $>$, or $=$.

Solution: Model A shows 2 flats and 5 rods: $200 + 50 = 250$. Model B shows 2 flats, 3 rods, and 9 cubes: $200 + 30 + 9 = 239$. The hundreds match (2 and 2), so compare the tens: 5 tens beats 3 tens, and the 9 extra ones cannot make up a missing ten.

250 > 239

7. Fill in the blanks for the number 407.

Solution: The 4 is in the hundreds place, the 0 is in the tens place (a placeholder — there are no loose tens), and the 7 is in the ones place. So $407 = \boxed{4}$ hundreds + $\boxed{0}$ tens + $\boxed{7}$ ones.

8. Challenge: a model shows 12 tens and 5 ones. What number is it? Explain the regrouping.

Solution: 12 tens is 120, and 5 ones is 5, so the model shows $120 + 5 = 125$.

125

What a good explanation contains: ten of the twelve tens are traded for one hundred flat, which leaves 2 rods. After the trade the model shows 1 hundred, 2 tens, and 5 ones — the same amount, regrouped.